NUFARM TRIFLURALIN 480 EC

Safety Data Sheet

Issue date: 1995 Revision date: 1/11/2022 Version: 2.0

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name	:	NUFARM TRIFLURALIN 480 SC
Product form	:	Emulsifiable concentrate
Type (Nufarm)	:	Herbicide
Country (Nufarm)	:	South Africa

1.2. Relevant identified uses of the substance or mixture and uses advised against

1.2.1. Relevant identified uses

Industrial/Professional use spec : Herbicide

1.2.2. Uses advised against

No additional information available

1.3. Details of the supplier of the safety data sheet

Registration holder :

Nufarm Agriculture (Pty) Ltd.

6 Eldo Close

Eldo Park

Eldoraigne , SOUTH AFRICA

T +27 (0)12 658 5869- F +27 (0)12 658 5869

www.nufarm.com/africa

1.4. Emergency telephone number

Emergency number : Griffon Poison Information Centre: +27 82 446 8946

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to the Regulations for Hazardous Chemical Agents, 2021

Carcinogenicity, Category 2	H351
Skin sensitization, Category 1	H317
Aspiration hazard, Category 1	H304
Specific target organ toxicity – Single exposure, Category 3, Narcosis	H336
Flammable liquids, Category 4	H227
Hazardous to the aquatic environment – Chronic Hazard, Category 1	H410

Full text of H-statements: see section 16

Adverse physicochemical, human health and environmental effects

Suspected of causing cancer. May cause an allergic skin reaction. May be fatal if swallowed and enters airways. May cause drowsiness or dizziness. Very toxic to aquatic life with long lasting effects. Combustible liquid.

2.2. Label elements

Labelling according to Globally Harmonized System of Classification and Labelling of Chemicals (GHS), 9th Revision (2021)

Hazard pictograms :

GHS08 GHS07 GHS09

Signal word : Danger

Contains : Trifluralin, Liquid Hydrocarbon

Hazard statements : H351 - Suspected of causing cancer.
H317 - May cause an allergic skin reaction.
H304 - May be fatal if swallowed and enters airways.
H336 - May cause drowsiness or dizziness.
H410 – Very toxic to aquatic life with long lasting effects.

Precautionary statements : P101 - If medical advice is needed, have product container or label at hand.
P102 - Keep out of reach of children.
P103 - Read label before use.
P203 - Obtain, read and follow all safety instructions before use.
P261 - Avoid breathing vapours or spray.
P272 - Contaminated work clothing should not be allowed out of the workplace.
P280 - Wear protective gloves, clothing, eye and face protection.
P271 - Use only outdoors or in a well-ventilated area.
P273 - Avoid release to the environment.
P308+P313 - IF exposed or concerned: Get medical advice/attention.
P302+P352 - IF ON SKIN: Wash with plenty of soap and water.
P333+P313 - If skin irritation or rash occurs: Get medical advice/attention.
P363 - Wash contaminated clothing before reuse.
P301+P316 - IF SWALLOWED: Get emergency medical help immediately.
P331 - Do NOT induce vomiting.
P304+P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P312 - Call a POISON CENTER or doctor/physician if you feel unwell.
P391 - Collect spillage.

SECTION 3: Composition/information on ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Name	Product identifier	%	Classification according to Regulation (EC) No. 1272/2008 [CLP]
Trifluralin	(CAS-No.) 1582-09-8	40-50 %	Skin Sen. 1, H317 Carc. 2, H351 Aquatic Acute 1, H400 Aquatic Chronic 1, H411
Emulsifier	N/A	< 10 %	Acute Tox. 4 (Inhalation), H332 Skin Irrit. 2, H315 Eye Dam. 1, H318 STOT SE 3, H335 Aquatic Chronic 3, H412
Liquid hydrocarbon	(CAS-No.) 64742-94-5	45 – 60 %	Asp. Tox. 1, H304 STOT SE 3, H336 Aquatic Chronic 1, H411

Full text of H-statements: see section 16

SECTION 4: First aid measures

4.1. Description of first aid measures

First-aid measures after inhalation : Remove affected person to fresh air until recovered. Apply CPR if there is no breathing and NO pulse.

First-aid measures after skin contact : Wash affected areas thoroughly with soap and water. Remove contaminated clothing and launder before re-use.

First-aid measures after eye contact : If in eyes, hold eyelids open and wash with copious amounts of water for at least 15 minutes.

First-aid measures after ingestion : If swallowed do NOT induce vomiting; seek medical advice immediately and show this container or label or contact the Poisons Information Centre. Make every effort to prevent vomit from entering the lungs by careful placement of the patient.

4.2. Most important symptoms and effects, both acute and delayed

No additional information available

4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically. If vomiting occurs, solvent present may cause pulmonary pneumonitis.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media : Water fog, foam, carbon dioxide or dry chemical.

Unsuitable extinguishing media : high volume water jet.

5.2. Special hazards arising from the substance or mixture

Fire hazard : Decomposition products may be a hazard to health.

Hazardous decomposition products in case of fire : If involved in a fire, it will emit hydrogen floride, oxides of nitrogen and possibly cyanides.

5.3. Advice for firefighters

Firefighting instructions : Exercise caution when fighting any chemical fire. Prevent fire fighting water from entering the environment 9drains or water bodies). Contain the extinguishing fluids by bunding.

Protection during firefighting : Wear a self contained breathing apparatus. Wear recommended personal protective equipment.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

General measures : Refer to chapter 8.

6.1.1. For non-emergency personnel

No additional information available

6.1.2. For emergency responders

For appropriate personal protective equipment (PPE), refer to section 8.

6.2. Environmental precautions

This product is a herbicide and spills can damage crops, pastures and desirable vegetation.
Prevent from entering drains, waterways or sewers.
Notify authorities if product enters sewers or public waters.

6.3. Methods and material for containment and cleaning up

For containment : Collect spillage.

Methods for cleaning up : Contain spill and absorb with clay, sand, soil or proprietary absorbent (such as vermiculite).

Other information : Collect spilled material and waste in sealable open-top type containers for disposal.

6.4. Reference to other sections

For further information refer to section 13. For further information refer to section 8: "Exposure controls/personal protection".

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Precautions for safe handling	:	Do not eat, drink or smoke when using this product.
Hygiene measures	:	Wash contaminated clothing before reuse. Always wash hands after handling the product.

7.2. Conditions for safe storage, including any incompatibilities

Storage conditions	:	Store in the closed, original container in a dry, well ventilated area out of direct sunlight. Do not store near oxidisers. Do not store below 5°C. Extended storage below 5°C can result in the formation of crystals on the bottom of the container. If crystallisation does occur, store the container on its side at room temperature and rock occassionally until crystals re-dissolve. Ensure any crystals are dissolved before adding to the spray tank. Keep away from food, drink and animal feeding stuffs. Keep out of the reach of children.

7.3. Specific end use(s)

Always read the label and any attached leaflet before use.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

8.1.1 National occupational exposure and biological limit values

No exposure standards have been set for this product or the active ingredients.
The manufacturer of the solvent has recommended an occupational exposure limit of 100 mg/m^3; 15ppm TWA, as total hydrocarbon.

8.1.2. Recommended monitoring procedures

No additional information available

8.1.3. Air contaminants formed

No additional information available

8.1.4. DNEL and PNEC

No additional information available

8.1.5. Control banding

No additional information available

8.2. Exposure controls

8.2.1. Appropriate engineering controls

Handle in well ventilated areas, generally natural ventilation is adequate.

8.2.2. Personal protection equipment

Personal protective equipment symbol(s):

8.2.2.1. Eye and face protection

Eye protection:
When opening the container, preparing spray and using the prepared spray wear face shield or goggles.

8.2.2.2. Skin protection

Skin and body protection:
When opening the container, preparing spray and using the prepared spray wear cotton overalls buttoned to the neck and wrist and a washable hat, elbow-length PVC gloves and face shield or goggles. Sensitive workers should use protective clothing. After use and before eating, drinking or smoking, wash hands, arms and face thoroughly with soap and water. After each day's use, wash contaminated clothing and safety equipment.

Hand protection:
When opening the container, preparing spray and using the prepared spray wear elbow-length PVC gloves

8.2.2.3. Respiratory protection
No additional information available

8.2.2.4. Thermal hazards
No additional information available

8.2.3. Environmental exposure controls

Environmental exposure controls:
Notify authorities if product enters sewers or public waters. Avoid release to the environment.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	:	Liquid
Colour	:	Bright orange, clear liquid
Odour	:	Solvent odour
Odour threshold	:	No data available
pH	:	7.5 - 8.5
pH solution concentration	:	No data available
Relative evaporation rate (butylacetate=1)	:	Not applicable
Melting point	:	Some crystallisation occurs between 0 and -7°C
Freezing point	:	No data available
Boiling point	:	183-210°C (for solvent)
Flash point	:	72.5°C
Auto-ignition temperature	:	443°C (solvent)
Decomposition temperature	:	No data available
Flammability (solid, gas)	:	Combustible liquid C1. (does not sustain combustion) AS1940. Combustible Liquid (GHS).
Vapour pressure	:	0.5 kPa @ 38°C for solvent, 13.7 mPa @ 25°C for trifluralin
Relative vapour density at 20 °C	:	4.8 (solvent)
Specific gravity	:	1.116
Solubility	:	Disperses in water.
Volatile Component	:	51% v/v
Partition coefficient n-octanol/water (Log Pow)	:	No data available
Viscosity, kinematic	:	No data available
Viscosity, dynamic	:	No data available
Explosive properties	:	No data available
Oxidising properties	:	No data available
Flammable limits Lower	:	0.9% (for solvent)
Flammable limits Upper	:	6% (for solvent)

9.2. Other information

No additional information available

SECTION 10: Stability and reactivity

10.1. Reactivity

Stable under recommended handling and storage conditions (see section 7).

10.2. Chemical stability

Stable under normal conditions. Do not store below 5°C.

10.3. Possibility of hazardous reactions

Violent reactions between this product and oxidising agents are possible.

10.4. Conditions to avoid

None under recommended storage and handling conditions (see section 7).

10.5. Incompatible materials

Prolonged reaction with water can cause slow decomposition and the formation of acid which may attack drums. If a part open drum is to be stored, ensure that no water has been added to the drum.

10.6. Hazardous polymerization products

Hazardous polymerisation is not possible.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

No harmful effects are expected if the precautions on the label and this SDS are followed.
Nufarm trifluralin (technical) is manufactured to comply with a maximum limit of 0.5ppm di-n-propylnitrosamine, generally, the levels are not detectable at a detection limit of 0.2ppm. The final product specification is set in accordance with the current FAO specification of max 0.5ppm di-n-propylnitrosamine.

Acute toxicity (oral)	:	LD_{50} (rat) >10,000 mg/kg for trifluralin LD50 (mouse) 5,000 mg/kg for trifluralin In 2 year feeding trials, rats receiving 2,000 mg/kg diet suffered no ill effects. In 2 year feeding trials, dogs receiving 1,000 mg/kg body weight suffered no ill effects.
Ingestion	:	The concentrate is of low toxicity if swallowed. However, swallowing of large amounts may cause injury, mainly due to the solvent. If aspirated into the lung, e.g. from vomiting, the presence of the solvent may result in chemical pneumonitis or other lung damage.
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Inhalation	:	When applying the product as a spray avoid breathing in spray mist. High vapour concentrations of the solvent while handling the concentrate are irritating to the eyes and the respiratory tract, may cause headaches and dizziness, can be anaesthetic and may have other central nervous system effects.
Skin	:	Prolonged contact with the concentrate can cause defatting of the skin and may result in dermatitis. May cause sensitisation by prolonged skin contact.
Eye	:	The concentrate may cause irritation of the eyes.
Carcinogenicity	:	The active ingredient trifluralin, and an impurity present in the solvent naphthalene have been found to cause tumors in laboratory animals and are classified as suspected human carcinogens.
Other Information	:	The Australian Acceptable Daily Intake (ADI) for trifluralin for a human is 0.02 mg/kg/day, set for the public for daily, lifetime exposure. This is based on the NOEL of 2.5 mg/kg/day, the level determined to show no effects during long term exposure for the most sensitive indicators and the most sensitive species. (Ref: Aust Govt Dept. of APVMA, 'ADI List', April 2020).

SECTION 12: Ecological information

12.1. Toxicity

Acute Toxicity - Fish	:	The following is data for the active ingredient, trifluralin. LC_{50} (96hr) for young rainbow trout is 0.088 mg/l. LC_{50} (96hr) for young bluegill sunfish is 0.089 mg/l.
Acute Toxicity - Daphnia	:	LC_{50} (48hr) for daphnia is 0.245 mg/l.

Acute Toxicity - Other Organisms	:	Birds: Not toxic to birds. LD_{50} for bobwhite quail is >2,000 mg/kg Bees: Not toxic to bees. LD_{50} >100 µg/bee
Hazardous to the aquatic environment, long–term (chronic)	:	Very toxic to aquatic life with long lasting effects.

12.2. Persistence and degradability

Trifluralin degrades in soil at a relatively moderate rate, about 85 - 90% of the material is lost in normal soil in 1/2 to 1 year.

12.3. Bioaccumulative potential

No additional information available

12.4. Mobility in soil

No additional information available

12.5. Results of PBT and vPvB assessment

No additional information available

12.6. Other adverse effects

Do not spray in high winds.
Do not contaminate dams, waterways or sewers with this product.

SECTION 13: Disposal considerations

13.1. Waste treatment methods

Sewage disposal recommendations	:	Disposal must be done according to official regulations. Do not discharge into drains.
Product/Packaging disposal recommendations	:	On site disposal of the concentrated product is not acceptable. Ideally, the product should be used for its intended purpose. If there is a need to dispose of the product, approach local authorities who hold periodic collections of unwanted chemicals. Do not re-use empty containers for any other purpose. Triple rinse containers before disposal. Add rinsings to the spray tank. Recycle: replace cap and return clean containers to recycler or designated collection point. Empty containers and product should not be burnt.

SECTION 14: Transport information

It is good practice not to transport agricultural chemical products with food, food related materials and animal feedstuffs.
Considered non dangerous for road and rail transport (in packaging, including IBCs not greater than 3000 Lt each) by the Australian Code for the Transport of Dangerous Goods by Road and Rail. Ref: ADG7; SP No. AU01.

14.1. UN number (Sea transport)
UN 3082
14.2. IMO proper shipping name
ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains trifluralin)
14.3. IMO Class
9
14.4. Packing group
III

14.5. Environmental hazards
Marine Pollutant: Yes

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. National regulations

South Africa

Refer to the Regulations for Hazardous Chemical Agents (2021).

15.2. Chemical safety assessment

No chemical safety assessment has been carried out

SECTION 16: Other information

Full text of H-statements:	
Carcinogenicity, Category 2	H351
Skin sensitization, Category 1	H317
Aspiration Hazard, 1	H304
Specific target organ toxicity – Single exposure, Category 3, Narcosis	H336
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H410	Very toxic to aquatic life with long lasting effects.

NUFARM SDS TEMPLATE

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.